RANGER EV POSITIVE TEMPERATURE COEFFICIENT (PTC) SWITCHING MODULE INSPECTION AND REPLACEMENT

AFFECTED VEHICLES: **CERTAIN 1999 AND 2000 MODEL YEAR RANGER EV**

INSPECTION

1. Open hood and locate part number/serial number tag on the back of the positive temperature coefficient (PTC) switching module. (The tag may be upside down, but is still visible). See Figure 1.

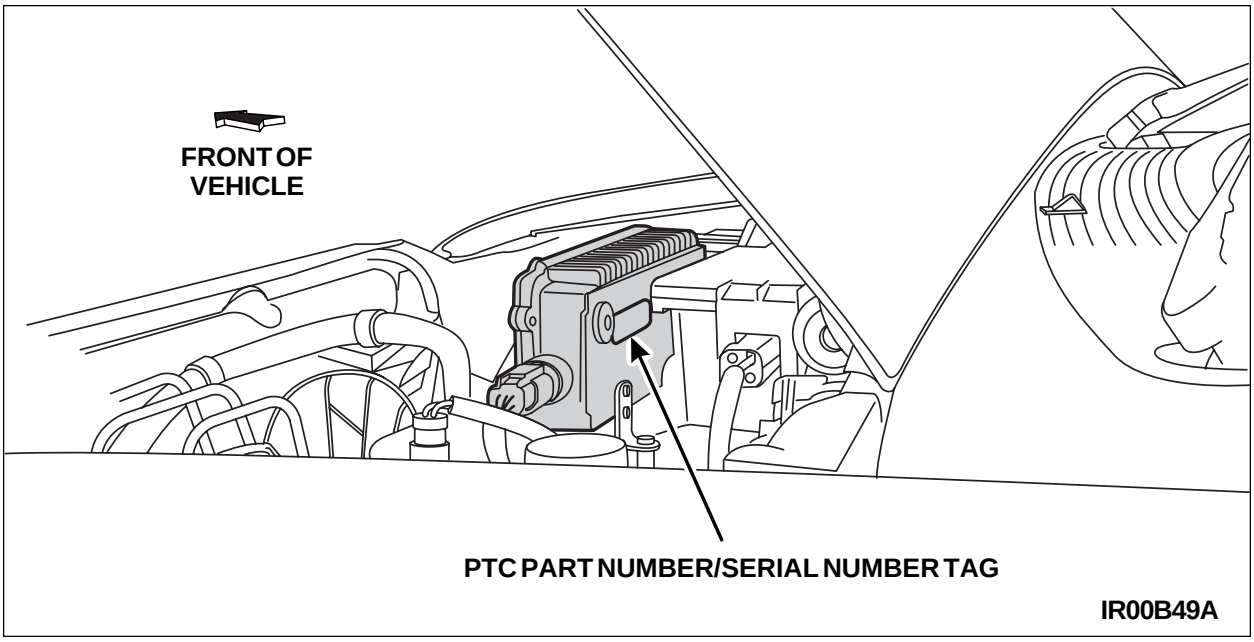


FIGURE 1

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2. Check the first three (3) digits of the serial number on the tag. See Figure 2. If it is GD7, the PTC must be replaced. Go to the Replacement procedure. If the serial number begins with anything other than GD7, no further action is required and the vehicle may be released.

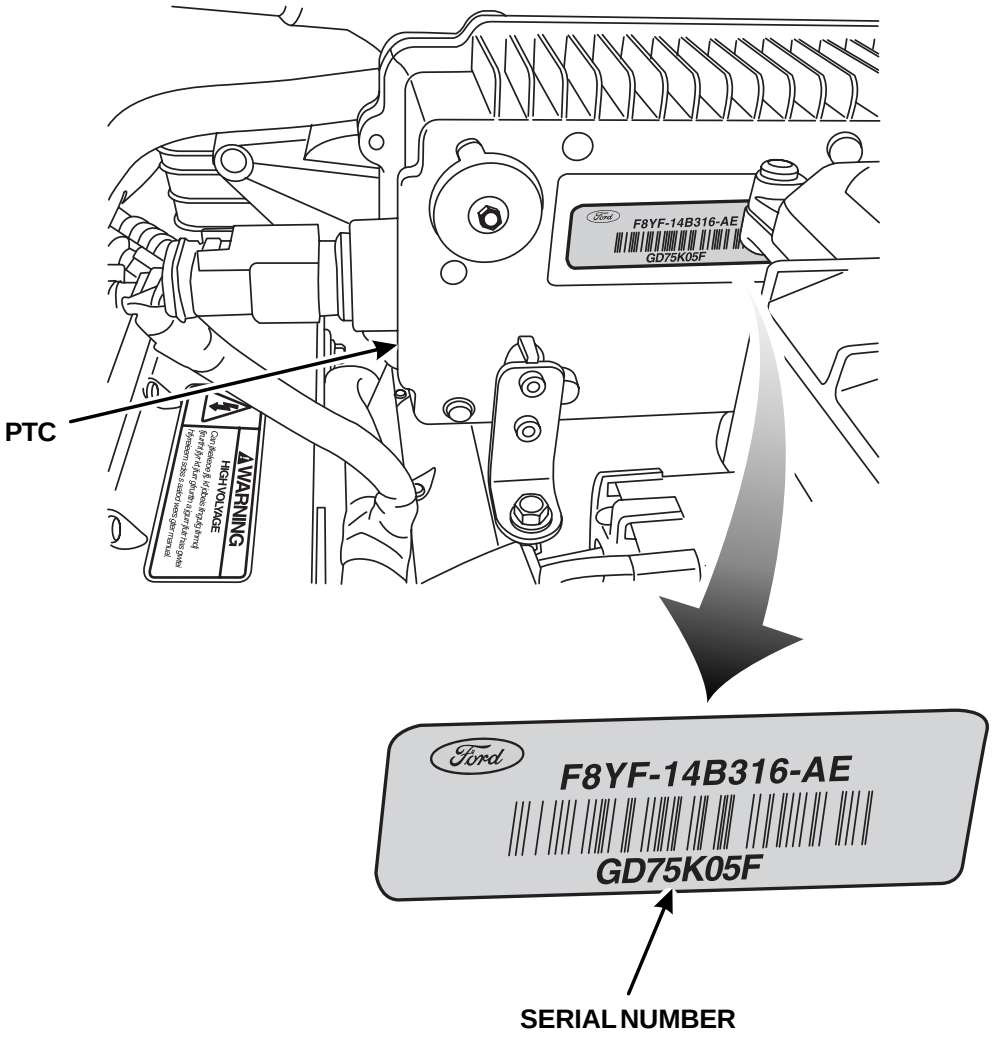


FIGURE 2

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REPLACEMENT

1. Record the radio stations.
2. Verify the vehicle is not connected to a charging station, then remove the ignition key from the ignition lock cylinder.
3. Disconnect the auxiliary battery negative cable.
4. Remove the high voltage power distribution box (HVPDB) cover. See Figure 3.

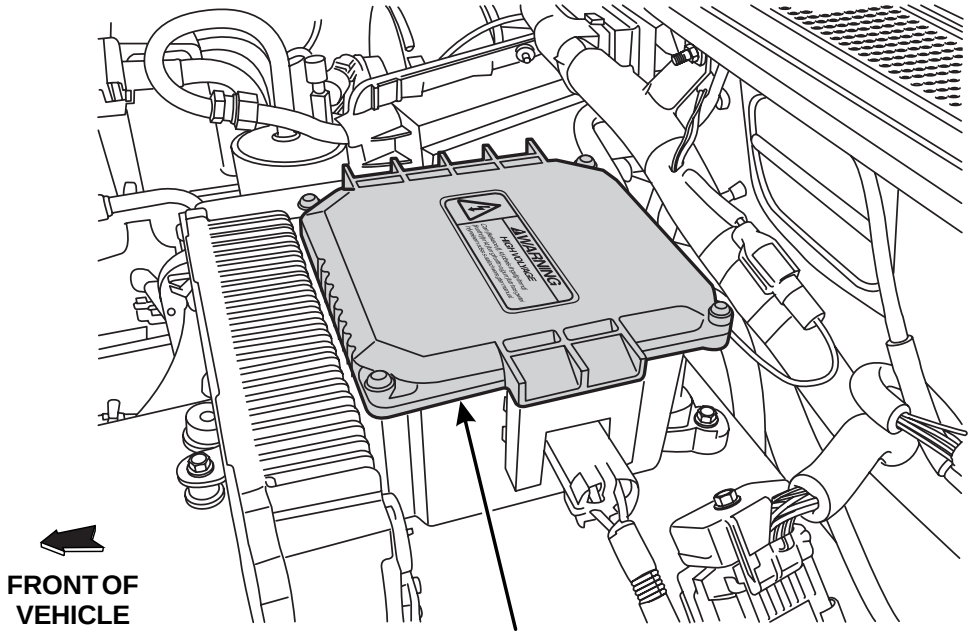


FIGURE 3

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5. Disconnect the PTC connectors in the following order (See Figure 4 for location and sequence):
   ① Driver-side connector (squeeze tab and pull)
   ② Passenger-side upper connector (pull up on red lock tab and disengage connector)
   ③ Passenger-side lower (harness) connector from the HVPDB. (Pull up on red lock tab and disengage connector) DO NOT unscrew harness connector from PTC.

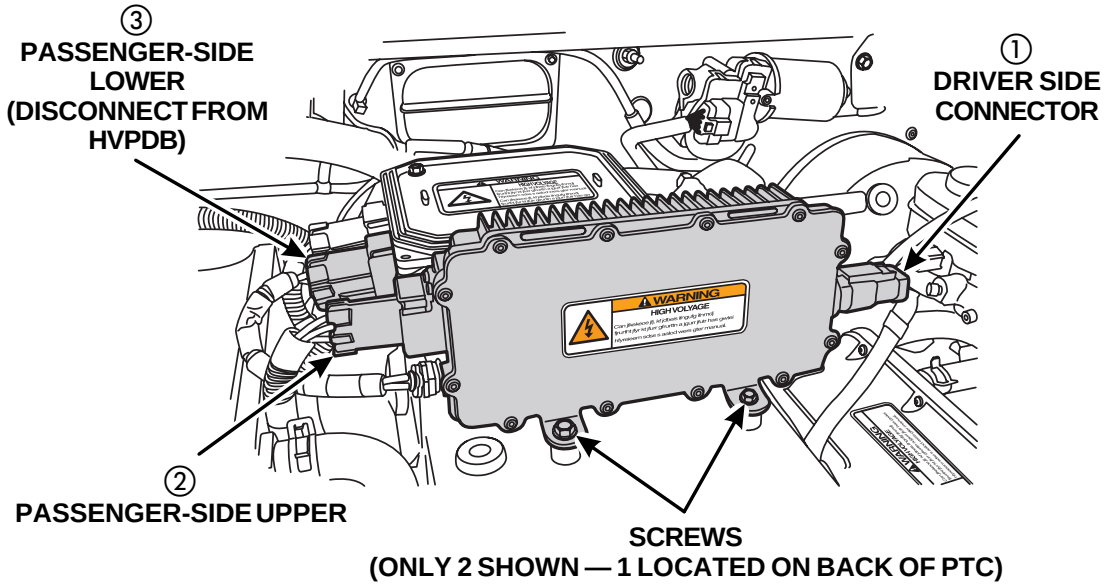


FIGURE 4

6. **CAUTION: Use hand tools only to remove the PTC fasteners to avoid damaging the threaded holes in the top of the charger.**

   Remove the three screws that attach the PTC and remove it from the vehicle. See Figure 4.

7. **CAUTION: Use hand tools only to install the PTC fasteners to avoid damaging the threaded holes in the top of the charger.**

   Install the ***new*** PTC. Tighten the retaining screws to 12 Nm (107 in-lb).

8. Connect the PTC connectors in reverse order, starting with the lower (harness) connector to the HVPDB, then the upper connector at the PTC, then the driver-side connector.
9. Install the HVPDB cover. Tighten the screws to 2 Nm (18 in-lb).
10. Connect the auxiliary battery negative cable.
11. Verify proper operation by turning the ignition to ON, then turn the climate control temperature control to the warmest setting and turn on the blower. Heat should begin to generate within a few seconds (10-30 seconds).
12. Reprogram the radio stations.

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